

Implementation Phases

heatwave_modelling_CHAP rework — full phase breakdown and current status

Status as of PR #1

Heatwave modelling pipeline (heatwave_modelling_CHAP) · Status report

Prepared for Adedo Lukmon · 10 September 2026

Overview

Nine phases take the repository from a single-file heat-index viewer to a modular, tested, ward-level heatwave-detection pipeline whose output is shaped as a covariate table usable downstream by CHAP. Phase A produced the review documents; Phases 0–8 are code implementation. Phases 0–2 are complete, verified against the live heatwave-508110 project, and shipped as pull request #1 (feature/heatwave-508110-phase-0-2 → main) on eHealthAfrica/heatwave_modelling_CHAP. Phases 3–8 have not started.

Phase	Name	Status
A	Review documents	Done
0	Packaging scaffold	Done
1	Config + auth consolidation	Done
2	Ward boundary + ERA5-Land switch	Done
3	Heat Index relocated + tested	Not started
4	Climatology + heatwave detection	Not started
5	Batch export + covariate table	Not started
6	Presentation layer rewrite	Not started
7	Documentation	Not started
8	Optional polish (not required)	Not started

Phase A — Review documents

Done

Produced the methodology and roadmap document pair in house style, for approval before any code changes.

- **Delivers:** [01_Heatwave_Methodology.docx](#), [02_Implementation_Roadmap.docx](#).

Phase 0 — Packaging scaffold

Done

Establishes the project as an installable package rather than a loose pair of scripts, and adds the test runner used from Phase 3 onward.

- **Delivers:** [pyproject.toml](#) (Python ≥3.11); [pytest](#) and [PyYAML](#) added to [requirements.txt](#).

Phase 1 — Config + auth consolidation

Done

Replaces the credential-resolution logic that was duplicated between `nigeria\_heat\_index.py` and `gee.py` with a single `init\_ee()`, still supporting the same three credential sources in the same priority order: Streamlit secrets, then the `EE\_SA\_JSON` environment variable, then a local `keys/service\_account.json`. `config.yaml` centralises the GCP project id, ward asset id, ERA5-Land band names, and the climatology constants fixed by the methodology document, so later phases read one source of truth instead of hardcoded values.

- **Delivers:** [heatwave/config.py](#), [config.yaml](#), [heatwave/auth.py](#). [gee.py](#) removed.

Bug found and fixed during implementation

Reading `st.secrets` when no secrets.toml exists at all raised `StreamlitSecretNotFoundError` instead of the intended not-configured fallback to the next credential source. This affected the original prototype too — it just never surfaced because it always ran with Streamlit Cloud secrets configured.

Phase 2 — Ward boundary + ERA5-Land switch

Done

Moves the app off the old, now-inaccessible ee-victoridakwo/Northern_Nigeria state-level boundary and the coarser ECMWF/ERA5/DAILY collection, onto the new nationwide ward-level asset and ECMWF/ERA5_LAND/DAILY_AGGR (~11.1 km resolution, daily record from 1950-01-02).

- **Delivers:** [heatwave/data/boundary.py](#), [heatwave/data/ingest.py](#).
- **Boundary asset:** [projects/heatwave-508110/assets/shp](#) — 4,841 wards nationwide (GRID3 schema: wardname, wardcode, lganame, statename).

Verified against the live project

Earth Engine authentication succeeds; the ward boundary loads with the expected feature count and properties; ERA5-Land ingestion returns the correct band names for a sample date range; the Streamlit app boots cleanly and serves over HTTP against real data.

Phases 1 and 2 were implemented together in one pass rather than sequentially: Phase 1's own checkpoint — the app still runs after this phase — was only achievable once the app pointed at the new asset and collection from Phase 2, since starting fresh meant no inherited access to the old project's boundary.

Phase 3 — Heat Index relocated + tested

Not started

Moves the existing relative-humidity approximation ($RH = 100 - 5 \times (T - D)$) and the NOAA/NWS Rothfus 9-coefficient Heat Index regression out of `nigeria\_heat\_index.py` into a tested module — relocated, not rewritten, per the methodology document.

- **Delivers:** [heatwave/science/heat_index.py](#), [tests/test_heat_index.py](#).

Phase 4 — Climatology + heatwave detection

Not started

The core new capability: computes each ward's 1991–2020 calendar-day 90th-percentile climatology (± 5 -day pooling window) and flags heatwave days and events (≥ 3 consecutive days exceeding the local percentile) via per-ward zonal reduction. The constants this phase needs are already in `config.yaml` from Phase 1.

- **Delivers:** [heatwave/science/climatology.py](#), [heatwave/science/heatwave.py](#), [heatwave/zonal.py](#), [tests/test_heatwave_detection.py](#).

Phase 5 — Batch export + covariate table

Not started

Produces the weekly time_period $\times$ location covariate table — this pipeline's actual deliverable to CHAP: time_period, location, heatwave_days, mean_heat_index, max_heat_index, heatwave_event_count.

- **Delivers:** [scripts/run_batch_export.py](#), [scripts/build_covariate_table.py](#), [heatwave/export.py](#), [tests/test_export.py](#).

Phase 6 — Presentation layer rewrite

Not started

Replaces the current live, per-date Earth Engine query pattern with a viewer that reads the precomputed covariate table from Phase 5, matching the pipeline's shift from a demo viewer to a covariate-producing pipeline.

- **Delivers:** [heatwave/app/streamlit_app.py](#). [nigeria_heat_index.py](#) retired.

Phase 7 — Documentation

Not started

- **Delivers:** [docs/METHODOLOGY.md](#), rewritten [README.md](#).

Phase 8 — Optional polish

Not started

- **Delivers:** [tests/test_config.py](#); optional CI workflow. Not required for the pipeline to function.

Verification checkpoints

- **After Phase 2 (passed)** — confirmed the uploaded ward asset loads with the expected ward count, and ERA5-Land images clip to it correctly.
- **After Phases 3, 4, 5** — pytest passes for Heat Index/RH, heatwave detection, and covariate export.
- **After Phase 5** — run the batch export on a short date range first; inspect the covariate CSV for one row per ward per week, plausible ranges, no missing periods.
- **After Phase 5/6** — sanity-check detected heatwave events against known historical hot spells for Northern Nigeria.